Chapter 8: Two-period Models

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8.1 An Overlapping Generations Model

- The model extends the deterministic OLG economy to a stochastic environment.
- The fundamental exogenous uncertainty is total factor productivity.
- Once technology and timing are specified, household and firm problems are derived.

8.1.1 Technology

- Aggregate production is Cobb-Douglas.
- Total factor productivity is positive and i.i.d.; no technological progress is imposed.

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha} \tag{8.1.1}$$

$$A_t \sim (\bar{A}, \sigma^2), \qquad A_t > 0. \tag{8.1.2}$$

- The distribution of A_t can be discrete or continuous; only mean and variance are initially required.

8.1.2 Timing

- At the beginning of period t, capital K_t is inherited from past decisions.
- Then productivity A_t is realized; firms choose factors and households choose consumption and saving.
- Wages and interest payments are only known with certainty at the end of the period.

$$\text{state in } t: \qquad (K_t, L_t, A_t).$$

- All endogenous variables are contingent on this state.

8.1.3 Firms

- Given the timing, firms bear no risk: factor prices are paid after A_t is known.
- Competitive factor payments equal marginal products.

$$w_t = p_t \frac{\partial Y_t}{\partial L_t}, \tag{8.1.3}$$

$$r_t = p_t \frac{\partial Y_t}{\partial K_t} = p_t A_t \alpha \left(\frac{L_t}{K_t} \right)^{1-\alpha}. \tag{8.1.4}$$

- The consumption good is the numeraire: $p_t \equiv 1$.
- All risk is borne by households through labor and capital income.

8.1.4 Intertemporal Utility Maximization

- Agents live for two periods and maximize expected utility.
- The expectation E_t is conditional on information available in period t.

$$\max \; E_t\{u(c_t) + \beta u(c_{t+1})\}. \tag{8.1.5}$$

$$w_t = c_t + s_t, \tag{8.1.6}$$

$$(1 + r_{t+1})s_t = c_{t+1}. \tag{8.1.7}$$

- Given first-period wage income, saving adjusts to satisfy the first-period budget constraint in realizations.

8.1.4 Intertemporal Utility Maximization: Savings

- Substitute the two budget constraints into utility.
- The only control variable is first-period saving s_t.

$$\max_{s_t} \; E_t\{u(w_t - s_t) + \beta u((1 + r_{t+1})s_t)\}.$$

- Because A_t is known before current consumption, $c_t = w_t - s_t$ is certain at the choice date.
- Future consumption remains uncertain because r_{t+1} depends on A_{t+1}.

$$\max_{s_t} \; u(w_t - s_t) + \beta E_t u((1 + r_{t+1})s_t).$$

8.1.4 Intertemporal Utility Maximization: FOC

- For a finite number of states $i = 1, \ldots, n$, let π_i be the probability of state i.

$$\max_{s_t} \left\{ u(w_t - s_t) + \beta \sum_{i=1}^{n} \pi_i u((1 + r_{i,t+1}) s_t) \right\}.$$

$$\begin{aligned} u'(w_t - s_t) &= \beta \sum_{i=1}^{n} \pi_i u'((1 + r_{i,t+1}) s_t)(1 + r_{i,t+1}) \\ &= \beta E_t u'((1 + r_{t+1}) s_t)(1 + r_{t+1}). \end{aligned} \tag{8.1.8}$$

- Optimality is ex ante: marginal utility today equals expected discounted marginal utility tomorrow adjusted by the return.

8.1.4 Intertemporal Utility Maximization: Cobb-Douglas

- With Cobb-Douglas preferences, expected utility is logarithmic across the two dates.

$$E_t\{\gamma \ln c_t + (1-\gamma)\ln c_{t+1}\}. \tag{8.1.9}$$

$$\begin{aligned} &\max_{s_t} E_t\{\gamma \ln(w_t - s_t) + (1-\gamma)\ln((1+r_{t+1})s_t)\} \\ &= \max_{s_t} \gamma \ln(w_t - s_t) + (1-\gamma)\left[\ln s_t + E_t \ln(1+r_{t+1})\right]. \end{aligned} \tag{8.1.10}$$

- The uncertainty term is additively separable from s_t.

8.1.4 Intertemporal Utility Maximization: CD Solution

- The first-order condition no longer contains the expectation term because of the logarithm.

$$\frac{\gamma}{w_t - s_t} = \frac{1-\gamma}{s_t}. \tag{8.1.11}$$

$$\begin{aligned} c_t &= \gamma w_t, \\ c_{t+1} &= (1-\gamma)(1+r_{t+1})w_t, \\ s_t &= (1-\gamma)w_t. \end{aligned} \tag{8.1.12}$$

- The solution has the same structure as the deterministic case, but c_{t+1} is uncertain from period t.

8.1.5 Aggregation and the Reduced Form: Consumption

- Let L newborns enter each period and set $L_t = L$.
- Young consumption aggregates from $c_t = \gamma w_t$ and competitive labor payments.

$$C_t^y = L\gamma w_t = \gamma(1-\alpha)Y_t.$$

- Old consumption depends on savings and the realized interest rate.

$$\begin{aligned} C_t^o &= L(1-\gamma)(1+r_t)w_{t-1} \\ &= (1-\gamma)(1+r_t)(1-\alpha)Y_{t-1}. \end{aligned}$$

8.1.5 Aggregation and the Reduced Form: Capital

- Next period's capital stock is aggregate saving by the young.
- Using the Cobb-Douglas saving rule gives one equation in one unknown.

$$\begin{aligned} K_{t+1} &= Ls_t \\ &= L(1-\gamma)w_t \\ &= (1-\gamma)(1-\alpha)Y_t \\ &= (1-\gamma)(1-\alpha)A_t K_t^{\alpha} L^{1-\alpha}. \end{aligned}$$

- Unlike the deterministic phase diagram, the K_{t+1} locus shifts with each realization of A_t.

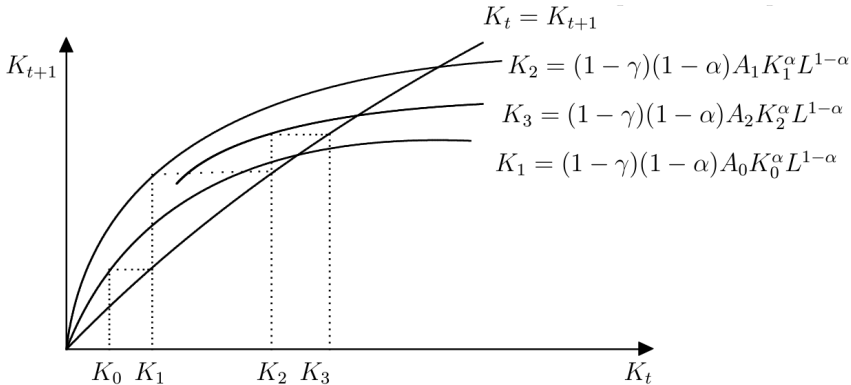


Figure 8.1.2 *Convergence towards a "stochastic steady state"*

8.1.6 Some Analytical Results: Basic Difference Equation

- Taking logs of the aggregate law of motion yields a stochastic difference equation.

$$\ln K_{t+1} = \ln\big((1-\gamma)(1-\alpha)\big) + \ln A_t + \alpha \ln K_t + (1-\alpha)\ln L_t.$$

$$\kappa_{t+1} = m_0 + \alpha\kappa_t + \nu_t, \qquad \nu_t \sim N(\mu_\nu, \sigma_\nu^2), \tag{8.1.13}$$

$$\kappa_t \equiv \ln K_t, \tag{8.1.14}$$

$$m_0 \equiv \ln[(1-\gamma)(1-\alpha)] + (1-\alpha)\ln L. \tag{8.1.15}$$

- Here $\nu_t \equiv \ln A_t$ is i.i.d.; write $\nu_t = \epsilon_t + \mu_\nu$ with $\epsilon_t \sim N(0, \sigma_\nu^2)$.

8.1.6 Some Analytical Results: Expected Value

- Solve the stochastic difference equation recursively from initial κ_0.

$$\kappa_t = \alpha^t \kappa_0 + (m_0 + \mu_\nu)\frac{1-\alpha^t}{1-\alpha} + \sum_{s=0}^{t-1} \alpha^{t-1-s}\epsilon_s. \tag{8.1.16}$$

$$E_0\kappa_t = \alpha^t \kappa_0 + (m_0 + \mu_\nu)\frac{1-\alpha^t}{1-\alpha}. \tag{8.1.17}$$

$$\lim_{t\to\infty} E_0\kappa_t = \frac{m_0 + \mu_\nu}{1-\alpha}. \tag{8.1.18}$$

- The expected log capital stock converges to the long-run mean of its stationary distribution.

8.1.6 Some Analytical Results: Variance

- Independence of the shocks implies that only the innovation term contributes to variance.

$$\begin{aligned}\mathrm{Var}(\kappa_t) &= \mathrm{Var}\left(\sum_{s=0}^{t-1} \alpha^{t-1-s}\epsilon_s\right) \\ &= \sum_{s=0}^{t-1} \alpha^{2(t-1-s)}\sigma_\nu^2 = \frac{1-\alpha^{2t}}{1-\alpha^2}\sigma_\nu^2. \end{aligned} \tag{8.1.19}$$

$$\lim_{t\to\infty} \mathrm{Var}(\kappa_t) = \frac{\sigma_\nu^2}{1-\alpha^2}. \tag{8.1.20}$$

- The stochastic steady state is a distribution, not a single deterministic capital stock.

8.1.6 Some Analytical Results: Fundamental Uncertainty

- Express the distribution of $\nu_t = \ln A_t$ in terms of the level of TFP.

$$\sigma_\nu^2 = \ln\left(1 + \left(\frac{\sigma_A}{\bar{A}}\right)^2\right), \tag{8.1.21}$$

$$\mu_\nu = \ln \bar{A} - \frac{1}{2}\sigma_\nu^2 = \ln \bar{A} - \frac{1}{2}\ln\left(1 + \left(\frac{\sigma_A}{\bar{A}}\right)^2\right). \tag{8.1.22}$$

$$\lim_{t\to\infty} E_0 \kappa_t = \frac{m_0 + \ln \bar{A} - \frac{1}{2}\ln\left(1 + (\sigma_A/\bar{A})^2\right)}{1-\alpha},$$

$$\lim_{t\to\infty} \mathrm{Var}(\kappa_t) = \frac{\ln\left(1 + (\sigma_A/\bar{A})^2\right)}{1-\alpha^2}.$$

- Higher TFP uncertainty lowers the long-run mean of log capital and raises its dispersion.

8.1.7 CRRA and CARA Utility Functions

- Arrow-Pratt absolute and relative risk aversion are

$$A(c) = -\frac{u''(c)}{u'(c)}, \qquad R(c) = -\frac{cu''(c)}{u'(c)}.$$

- For CRRA/CES utility,

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \qquad A(c) = \frac{\sigma}{c}, \qquad R(c) = \sigma.$$

- For CARA utility,

$$u(c) = -e^{-\sigma c}, \qquad A(c) = \sigma.$$

- CRRA links risk aversion and intertemporal substitution through the same parameter σ.

8.2 Risk-averse and Risk-neutral Households

- Individuals can lend to each other at a riskless endogenous interest rate r.
- There is also a normal asset paying uncertain return r_{t+1}.
- There are two types: risk-averse and risk-neutral, $i = a, n$.
- Heterogeneity is crucial: with identical agents and identical income streams, no loans would be traded in equilibrium.
- The analysis is kept in partial equilibrium.

8.2 Risk-averse and Risk-neutral Households: Budget Constraints

- Household i chooses saving s_t^i and portfolio share θ^i in the risky asset.

$$w_t = c_t^i + s_t^i, \tag{8.2.1}$$
$$c_{t+1}^i = s_t^i(1 + r^i), \tag{8.2.2}$$
$$r^i = \theta^i r_{t+1} + (1 - \theta^i)r. \tag{8.2.3}$$

- The share θ^i denotes the fraction of wealth held in the risky asset.
- Second-period consumption is entirely financed by capital income.

8.2 Risk-averse and Risk-neutral Households: Household Problem

- Substitution gives an unconstrained problem with two controls: s_t^i and θ^i.

$$U_t^i = u(w_t - s_t^i) + \beta E_t u\left((1 + r^i)s_t^i\right) \quad \max_{s_t^i, \theta^i}. \tag{8.2.4}$$

$$u'(c_t^i) = \beta E_t \left\{u'(c_{t+1}^i)(1 + r^i)\right\}, \tag{8.2.5}$$

$$E_t u'(c_{t+1}^i)[r_{t+1} - r] = 0. \tag{8.2.6}$$

- The first condition governs intertemporal consumption; the second governs asset allocation.

8.2 Risk-averse and Risk-neutral Households: Interpretation

- Rewrite the consumption FOC as a stochastic Euler equation.

$$E_t\left\{\frac{\beta u'(c_{t+1}^i)}{u'(c_t^i)}(1+r^i)\right\}=1. \tag{8.2.7}$$

- Current and future marginal utilities are equated in expected present-value terms.
- The portfolio FOC says expected utility-weighted returns from the risky and riskless assets are equal.

$$E_t u'(c_{t+1}^i) r_{t+1} = r E_t u'(c_{t+1}^i).$$

8.2 Risk-averse and Risk-neutral Households: Risk-neutral

- With linear utility, marginal utility is constant.

$$1 = \beta E\left[1 + \theta^n r_{t+1} + (1-\theta^n) r\right], \tag{8.2.8}$$

$$E r_{t+1} = r. \tag{8.2.9}$$

- Combining the investment FOC with the consumption FOC pins down the loan rate.

$$1 = \beta(1+r) \quad \Longleftrightarrow \quad r = \rho, \qquad \beta = (1+\rho)^{-1}. \tag{8.2.10}$$

- An interior solution requires $E r_{t+1} = \rho$.

8.2 Risk-averse and Risk-neutral Households: Risk-averse Saving

- Let risk-averse households have logarithmic utility.
- Insert $r^a = \theta^a r_{t+1} + (1-\theta^a) r$ into the Euler equation.

$$\begin{aligned}\frac{1}{c_t^a} &= \beta E\left\{\frac{1}{c_{t+1}^a}[1+\theta^a r_{t+1} + (1-\theta^a) r]\right\} \\ &= \beta E \frac{1}{s_t^a} = \beta \frac{1}{s_t^a}.\end{aligned} \tag{8.2.11}$$

$$\begin{aligned}c_t^a &= \frac{1}{1+\beta} w_t, \\ s_t^a &= \frac{\beta}{1+\beta} w_t, \\ c_{t+1}^a &= \frac{\beta}{1+\beta}[1+\theta^a r_{t+1} + (1-\theta^a) r] w_t.\end{aligned} \tag{8.2.12}$$

8.2 Risk-averse and Risk-neutral Households: Portfolio Derivative

- Differentiate the objective with respect to the risky-asset share.
- Insert the logarithmic-consumption solution.

$$\begin{aligned}\frac{d}{d\theta}U_t^a &= \beta E\frac{1}{c_{t+1}^a}(r_{t+1}-r)\\ &= (1+\beta)E\frac{r_{t+1}-r}{1+\theta^a r_{t+1}+(1-\theta^a)r}\\ &= (1+\beta)E\frac{r_{t+1}-r}{1+\theta^a(r_{t+1}-r)+r}\\ &\equiv (1+\beta)E\frac{X}{\phi+\theta^a X}. \end{aligned} \tag{8.2.13}$$

- Definitions: $X \equiv r_{t+1}-r$ and $\phi \equiv 1+r$.

8.2 Risk-averse and Risk-neutral Households: Who Owns What?

- The derivative of the investment FOC is negative for all θ^a.

$$\frac{d}{d\theta^a} E \frac{X}{\phi + \theta^a X} = E\left(-\frac{X^2}{(\phi + \theta^a X)^2}\right) < 0 \qquad \forall \theta^a.$$

- Using $Er_{t+1} = r$, we have $EX = 0$; hence $\theta^a = 0$ satisfies the FOC.

$$\theta^a = 0 \quad \Rightarrow \quad E\frac{X}{\phi + \theta^a X} = EX = 0,$$

$$E\frac{X}{\phi + \theta^a X} = 0 \quad \Longleftrightarrow \quad \theta^a = 0.$$

- Risk-averse individuals allocate all savings to loans; risk-neutral individuals bear all risk.

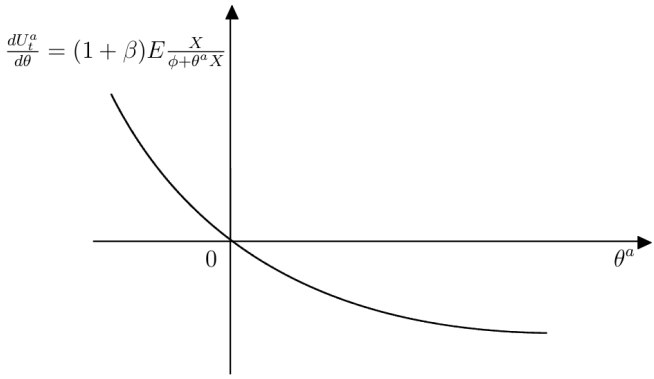


Figure 8.2.1 *The first-order condition (8.2.13) for the share θ^a of savings held in the risky asset*

8.3.1 The Value of an Asset: The Individual

- The question is what an individual pays at $t = 0$ for an asset worth uncertain p_{j1} at $t = 1$.
- Preferences are

$$E\{u(c_0) + \beta u(c_1)\}.$$

- First-period income is allocated to consumption, risky assets j, and a riskless bond.

$$w = c_0 + \sum_{j=1}^{n} m_j p_{j0} + b,$$

$$c_1 = \sum_{j=1}^{n} m_j p_{j1} + (1 + r)b.$$

8.3.1 The Value of an Asset: Maximization

- Substitute the budget constraints into the objective.
- Controls are current consumption c_0 and the number of shares m_j.

$$\max_{c_0, m_j} u(c_0) + \beta E u \left((1+r) \left(w - c_0 - \sum_j m_j p_{j0} \right) + \sum_j m_j p_{j1} \right).$$

- Uncertainty enters only through second-period consumption.

8.3.1 The Value of an Asset: Pricing Equation

- First-order conditions for consumption and asset holdings imply asset prices.

$$u'(c_0) = (1+r)\beta E u'(c_1).$$

$$\beta E\left[u'(c_1)(-(1+r)p_{j0} + p_{j1})\right] = 0 \quad \Longleftrightarrow \quad E[u'(c_1)p_{j1}] = (1+r)p_{j0}Eu'(c_1).$$

$$p_{j0} = \frac{1}{1+r} E\left[\frac{u'(c_1)}{Eu'(c_1)} p_{j1}\right]. \tag{8.3.1}$$

- The future asset price is discounted and weighted by marginal utility.

8.3.2 The Value of a Contingent Claim

- A contingent claim pays only under specified outcomes.
- Let its payoff be $g(p_{j1})$ and its current value be $v(p_{j0})$.

$$v(p_{j0}) = \frac{1}{1+r} E\left[\frac{u'(c_1)}{Eu'(c_1)} g(p_{j1})\right]. \tag{8.3.2}$$

- If all agents are risk neutral, $u'(c_1)$ is constant, so marginal-utility weights disappear.

$$v(p_{j0}) = \frac{1}{1+r} Eg(p_{j1}). \tag{8.3.3}$$

8.3.3 Risk-neutral Valuation

- Risk-neutral valuation asks when (8.3.3) holds even with risk-averse households.
- Drop the asset index and let p_1 have density $f(p_1;\mu)$, where $\mu \equiv Ep_1$.

$$v(p_0) = \frac{1}{1+r}\int g(p_1)f(p_1;\mu)\,dp_1, \qquad \mu = (1+r)p_0.$$

- Under risk aversion, the same expression becomes

$$v(p_0) = \frac{1}{1+r}\int \frac{u'(c_1)}{Eu'(c_1)}g(p_1)f(p_1;\mu)\,dp_1.$$

- A RNVR holds if the marginal-utility term can be absorbed into the density:

$$\frac{u'(c_1)}{Eu'(c_1)} = \frac{f(p_1)}{f(p_1;\mu)}.$$

8.4 Natural Volatility I

- Natural volatility views long-run growth and short-run fluctuations as jointly determined by real economic forces.
- Both growth and fluctuations stem from the introduction of new technologies.
- No exogenous shocks occur in this approach.
- This differs from RBC models, sunspot models, and endogenous-growth models with exogenous disturbances.
- The section presents a simple model to isolate the basic intuition.

8.4.1 The Basic Idea

- Productivity does not grow smoothly; it follows a step function.
- Periods of no productivity change are interrupted by discrete technology jumps.
- After a jump, investment returns rise and growth jumps upward; growth then gradually falls.

$$\text{long-run growth} + \text{short-run fluctuations} \quad \Longleftarrow \quad \text{step-wise productivity improvements.}$$

- The step function must arise from deeper economic mechanisms, not be simply imposed.

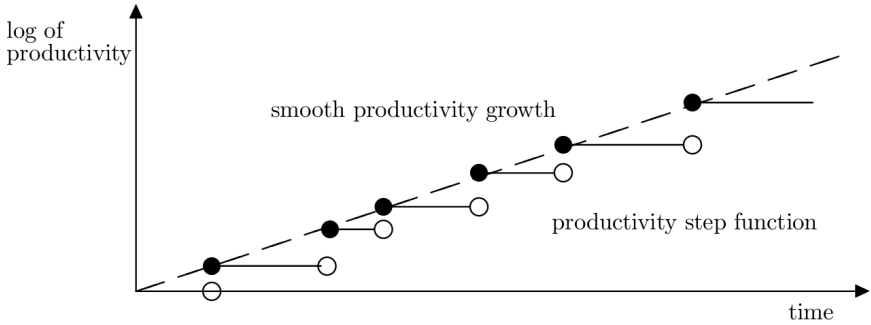


Figure 8.4.1 *Smooth productivity growth in balanced growth models (dashed line) and step-wise productivity growth in models of natural volatility*

8.4.2 A Simple Stochastic Model: Technologies

- Output is produced with Cobb-Douglas technology and constant labor input.

$$Y_t = A_t K_t^{\alpha} L^{1-\alpha}.$$

- Capital accumulates according to

$$K_{t+1} = (1-\delta)K_t + I_t.$$

- TFP evolves through stochastic jumps.

$$A_{t+1} = (1+q_t)A_t, \tag{8.4.1}$$

$$q_t = \begin{cases} \bar{q}, & \text{with probability } p_t, \\ 0, & \text{with probability } 1-p_t, \end{cases} \tag{8.4.2}$$

8.4.2 A Simple Stochastic Model: R&D

- The probability of a technology jump depends on R&D resources.

$$p_t = p(R_t), \qquad 0 \leq p(R_t) \leq 1. \tag{8.4.3}$$

- The key difference from RBC-type approaches is that the probability of technological progress is endogenous.
- Growth and fluctuations both stem from jumps in q_t.

source of growth = source of fluctuations = technology jumps.

8.4.2 A Simple Stochastic Model: Households

- The economy obeys the per-period resource constraint.

$$C_t + I_t + R_t = Y_t.$$

- Optimal behavior is summarized by constant shares of output devoted to consumption and R&D.

$$C_t = s_C Y_t, \qquad R_t = s_R Y_t.$$

- The saving rule could arise from a two-period or infinite-horizon maximization problem under parameter restrictions.

8.4.3 Equilibrium

- Equilibrium capital accumulation follows from the resource constraint and constant expenditure shares.

$$\begin{aligned} K_{t+1} &= (1-\delta)K_t + Y_t - R_t - C_t \\ &= (1-\delta)K_t + (1 - s_R - s_C)Y_t. \end{aligned}$$

- The probability of a technology jump is determined by R&D investment.

$$p_t = p(s_R Y_t).$$

- Given A_0, if $q_t = 0$ for several periods, capital approaches a temporary steady state.

8.4.3 Equilibrium: Temporary Steady State

- With no technology jump, $A_t = A_0$ and the temporary steady state is defined by $K_{t+1} = K_t \equiv K$.

$$\begin{aligned} \delta K &= (1 - s_R - s_C) A_0 K^{\alpha} L^{1-\alpha}, \\ K^{1-\alpha} &= \frac{1 - s_R - s_C}{\delta} A_0 L^{1-\alpha}. \end{aligned} \tag{8.4.4}$$

- If a technology jump occurs in period 12, then $A_{12} = (1 + \bar{q}) A_0$.
- The K_{t+1} schedule shifts upward, raising the temporary steady state.
- Capital repeatedly approaches a moving target: a Sisyphus-type economy.

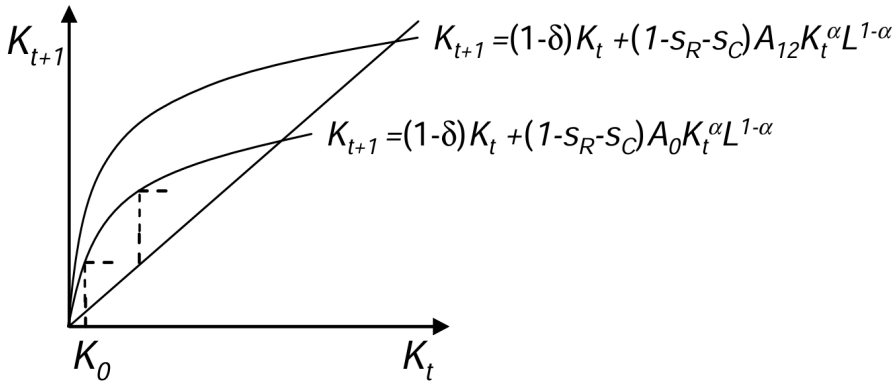


Figure 8.4.2 *The Sisyphus economy - convergence to a temporary steady state*